

Dominance-Solvable Games

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What is the evidence on reasoning depths?

- Avoiding dominated strategies is a basic principle in decision making
- Assuming others avoid dominated strategies is a model of how others behave.
- Assuming others assume you avoid dominated strategies is a model of how others model you.
- Dominance solvable games may require many iterations of this sort of abstraction assumption before they result in a unique equilibrium.

- Knowing how many steps of iterated reasoning people use is essential for making accurate predictions or giving effective advice

Dominance solvable games

- Strategy A *strictly dominates* strategy B if the payoff from choosing A is higher than the payoff from B for all combinations of the other players' strategies.
- Strategy A *weakly dominates* strategy B if the payoff from choosing A is sometimes higher and never lower than the payoff from B for all combinations of the other players' strategies
- Games in which a process of iteratively deleting dominated strategies leads to a unique equilibrium are called *dominance solvable*.

Evidence on the Fact of Rationality

- No conflict between individual rationality and collective rationality (Pareto efficiency)
- Subjects usually avoid strictly dominated strategies at frequencies that are higher than 90 percent.

	T	M	B	% play
T	320	440	500	1
M	420	600	660	94
B	180	360	420	5

Figure 1: Cooper et al. (1990) Game 1

Conflict between individual and collective rationality

- With a conflict between individual and collective rationality (Pareto efficiency) the rate at which subjects avoid dominated strategies is very sensitive to measurement abstraction assumptions
- Cooper et al. (GEB, 1996)
 - turnpike protocol
 - 40 subjects each of whom played 20 games and never meet anyone who had met someone ... who had met them before.
 - Think of Texas Avenue with 20 people who are north of university and start walking south and 20 people who are south of university and start walking north.

Cooper et al. (GEB, 1996)

	T	B	% play
T	350	1000	78
B	0	800	22

Figure 3: Cooper et al. (1996) Prisoners' Dilemma

Van Huyck, Wildenthal, and Battalio (2002)

- Used evolutionary protocol

	a_1	a_2	a_3	a_4	a_5	% play period 1	% play period 55
a_1	7,7	0,0	0,0	0,0	0,11	28	2
a_2	0,0	6,6	0,0	0,0	0,10	0	0
a_3	0,0	0,0	5,5	0,0	0,9	7	0
a_4	0,0	0,0	0,0	4,4	0,8	5	0
a_5	11,0	10,0	9,0	8,0	3,3	60	98

Van Huyck *et al.* Game S



Strategies undominated in pure strategies, but strictly dominated by mixed strategies

- No conflict with efficiency

	R	S	P	D	matrix cents	bimatrix cents	bimatrix probability
R	15 15	0 75	75 0	25 5	21	24	24
S	75 0	15 15	0 75	25 5	24	27	30
P	0 75	75 0	15 15	25 5	24	23	27
D	5 25	5 25	5 25	0 0	31	26	20

Trusting the Rationality of Others

- Ignore the distinction between one round of elimination of weakly dominated strategies and one step of backwards induction
- Beard and Beil's (1994) –

	l	r	Extensive Form %
L	9.75,3.00	9.75,3.00	66
R	3.00,4.75	10.00,5.00	34
Extensive Form %	17	83	

Beard and Beil (1994)

Table 5.2. Payoff treatments and results in Beard and Beil

Treatment	Payoffs from			Frequency of		Number of pairs	Threshold $p(r R)$
	(L, l)	(R, l)	(R, r)	L	$r R$		
1 (baseline)	(9.75,3)	(3,4.75)	(10,5)	0.66	0.83	35	.97
2 (less risk)	(9,·)	(·,·)	(·,·)	0.65	1.00	31	.85
3 (even less risk)	(7,·)	(·,·)	(·,·)	0.20	1.00	25	.57
4 (more assurance)	(·,·)	(·,3)	(·,·)	0.47	1.00	32	.97
5 (more resentment)	(·,6)	(·,·)	(·,·)	0.86	1.00	21	.97
6 (less risk, more reciprocity)	(·,5)	(5,9.75)	(·,10)	0.31	1.00	26	.95
7 (1/6 payoff)	(58.5,18)	(18,28.5)	(60,30)	0.67	1.00	30	.97

Source: Beard and Beil (1994).

Note: (·,·) indicates the payoffs are the same as those in the baseline case.

Beard and Beil (1994)

- Behavior is sensitive to the incentive to optimize and equity considerations
- Majority doesn't trust the rationality of opponent when an error is costly.

Goeree and Holt (1999)

Table 5.3. *Goeree and Holt's credible threat games*

Condition	Number of pairs	Threshold $p(r R)$	Payoffs			Frequency of	
			(L)	(R,l)	(R,r)	L	r R
Baseline 1	25	.33	(70,60)	(60,10)	(90,50)	0.12	1.00
Lower assurance	25	.33	(70,60)	(60,48)	(90,50)	0.32	0.53
Baseline 2	15	.85	(80,50)	(20,10)	(90,70)	0.13	1.00
Lower assurance	25	.85	(80,50)	(20,68)	(90,70)	0.52	0.75
Very low assurance	25	.85	(400,250)	(100,348)	(450,350)	0.80	0.80

Source: Goeree and Holt (1999).

Tree-Matrix Differences

- Schotter, Weigelt, and Wilson (1994) compare iterated dominance and subgame perfection.

Table 5.4. *Games 1M and 1S of Schotter et al.*

Player 1	Player 2		Actual frequency
	l	r	
<i>Normal form (1M)</i>			
L	4,4	4,4	(0.57)
R	0,1	6,3	(0.43)
Frequency	(0.20)	(0.80)	
<i>Sequential form (1S)</i>			
L	4,4		(0.08)
	l	r	
R	0,1	6,3	(0.92)
Frequency	(0.02)	(0.98)	

Source: Schotter, Weigelt, and Wilson (1994).

Schotter et al. (1994)

- A large majority of player 2s do obey dominance
 - 80 percent in the bimatrix form
 - 98 percent in the tree form
- But a majority are willing to trust rationality only in the tree form
- Matrix-tree “presentation effect”
- Van Huyck and Cooper (2003) find a similar difference that they attribute to a preference for giving the other player a choice.

Beauty Contest Games

Professional investment may be likened to those newspaper competitions in which the competitors have to pick out the six prettiest faces from a hundred photographs, the prize being awarded to the competitor whose choice most nearly corresponds to the average preferences of the competitors as a whole; so that each competitor has to pick, not those faces which he himself finds prettiest, but those which he thinks likeliest to catch the fancy of the other competitors, all of whom are looking at the problem from the same point of view. It is not a case of choosing those which, to the best of one's judgment, are really the prettiest, nor even those which average opinion genuinely thinks the prettiest. We have reached the third degree where we devote our intelligences to anticipating what average opinion expects the average opinion to be. And there are some, I believe, who practice the fourth, fifth and higher degrees.

Keynes (1936)



P-Beauty Contest

- Each of N players i choose a number x_i in the interval $[0,100]$ simultaneously.
- A multiple p of the average of their numbers defines a target number.
- The player whose number is closest to the target number wins a fixed prize.



Nagel and Duffy (EJ, 1997)

	Interval	$\frac{1}{2}$ - Median Game % Play	$\frac{1}{2}$ - Mean Game % Play
$\sim R$	(50, 100]	5	10
$\sim(ek)R - \sim R$	(25, 50]	27	30
$\sim(ek)^2R - \sim(ek)R$	(12.5, 25]	45	45
$\sim(ek)^3R - \sim(ek)^2R$	(6.25, 12.5]	16	10
$S - \sim(ek)^3R$	[0, 6.25]	8	5



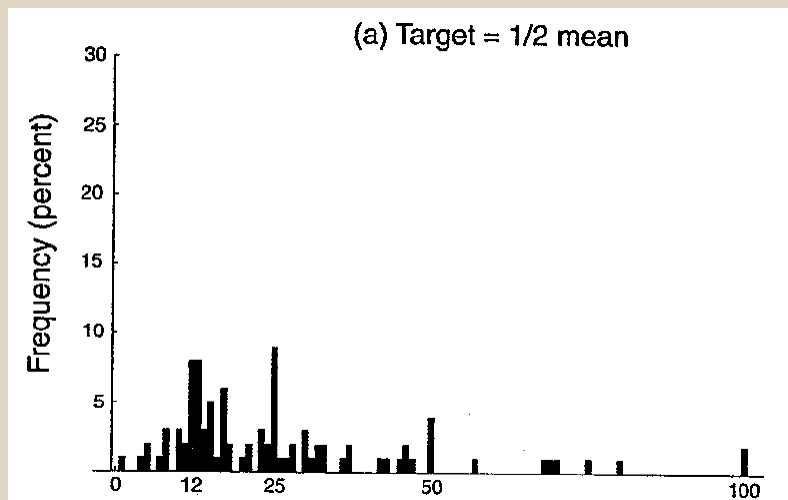
Van Huyck, Wildenthal, and Battalio (2002)



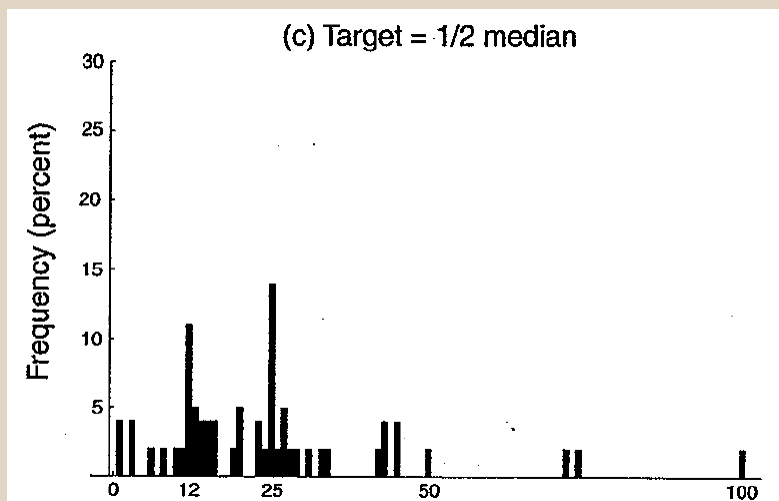
	a_1	a_2	a_3	a_4	a_5		Period 1 % play	Period 55 % play
a_1	7,7	0,11	0,0	0,0	0,0	$\sim R$	12	2
a_2	11,0	6,6	0,10	0,0	0,0	$\sim(ek)R - \sim R$	20	5
a_3	0,0	10,0	5,5	0,9	0,0	$\sim(ek)^2R - \sim(ek)R$	53	18
a_4	0,0	0,0	9,0	4,4	0,8	$\sim(ek)^3R - \sim(ek)^2R$	10	5
a_5	0,0	0,0	0,0	8,0	3,3	$S - \sim(ek)^3R$	5	70

Game I

Nagel (1999)



Nagel (1999)



Level-k Thinking

- Peaks at 50, 25, 12 suggest a model of iterative reasoning first developed by Stahl and Wilson (1995).
- Level-0 players- Assume a fraction w_0 of subjects just choose a number randomly from a normal distribution.
- Level-1 players – A fraction w_1 of subjects think all others are level-0 players and choose a noisy best response.
- Level-2 players – A fraction w_2 of subjects think all others are one level below them and choose a noisy best response.

Ho, Camerer, and Weigelt (1998)



Table 5.7. *Estimated fractions ω_k of level- k types in beauty contest games*

Estimate	Ho, Camerer, and Weigelt (1998)		Nagel (1995)	
	$p > 1$ games	$p < 1$ games	$p = 1/2$	$p = 2/3$
ω_0	0.22	0.16	0.16 (0.24)	0.28 (0.13)
ω_1	0.31	0.21	0.38 (0.30)	0.34 (0.44)
ω_2	0.13	0.13	0.47 (0.41)	0.37 (0.39)
ω_3	0.34	0.50	0.00 (0.06)	0.00 (0.03)

Note: Numbers in parentheses indicate Nagel's original estimates.

Camerer (unpublished)

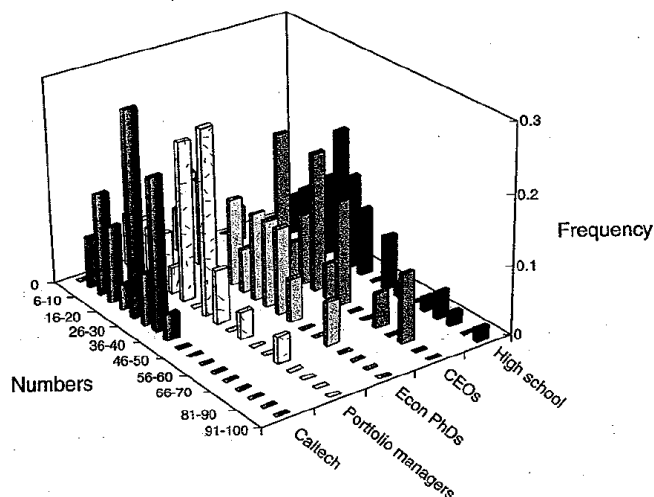
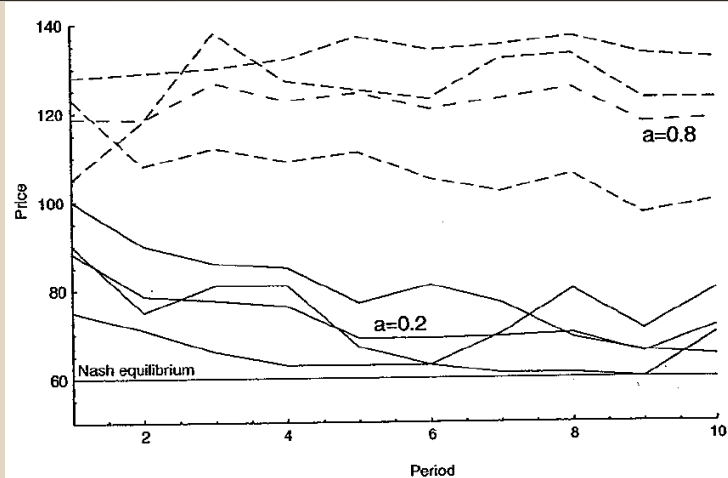


Figure 5.2. *Beauty contest choices from various exotic subject pools. Source: Camerer (unpublished).*

Price Competition: Capra et al. (2002)

- Two firms choose prices p_1 and p_2 from the interval $[\$0.60, \$1.60]$.
- Assume the low price firm is indexed by 1
- Then the low price firm earns p_1 and the high price firm earns $a p_1$. $0 < a < 1$
- If the prices are equal, they each earn $(1+a)p_1/2$
- The unique Nash equilibrium is for each firm to choose the lowest price.
- However, when a is close to 1 the incentive to undercut price is small, while when a is close to 0 the incentive to undercut price is large.

Price Competition: Capra et al. (2002)



Price Competition: Capra et al. (2002)

- Behavior depends on the incentive to be the low price seller, which is the sort of sensitivity predicted by Logit Equilibria.
- When a is low there is a large cost to being the high price seller and behavior is close to the Nash Equilibrium.
- But when a is high and there is little incentive to give a best response behavior is far from Nash equilibrium.

Cooper et al. (1990)

- Dominated strategies can influence equilibrium selection, which is evidence against everyone knowing the players are rational.

	T	M	B	% play period 1	% play period 22
T	350	350	1000	40	90
M	250	550	0	20	10
B	0	0	600	40	0

Figure 13: Cooper et al. (1990) Game 3

	T	M	B	% play period 1	% play period 22
T	350	350	700	10	0
M	250	550	1000	90	90
B	0	0	600	0	10

Figure 14: Cooper et al. (1990) Game 5

Cooper et al. (1993)

- BT requires the column player to be rational and own money motivated and to know the row player is rational and own money motivated.

	T	M	% play bimatrix	% play Extensive
T	300,300	300,300	42	20
M	0,0	200,600	5	2
B	600,200	0	53	78
Bimatrix	64	36		
Extensive	92	8		

Figure 15: Cooper et al. (1990) Game 3

Summary

- Most subjects avoid the use of strictly dominated strategies
 - My rule-of-thumb when designing an experiment is that about 5 percent of subjects in an experiment are noise players.
 - The use of strictly dominated strategies is sensitive to measurement abstraction assumptions when
 - Efficiency conflicts with dominance
 - Dominance relies on mixed strategies

Summary

- Subjects don't attribute as much rationality to their opponents as they appear to have, that is, it seems difficult to make everyone believe that almost everyone is rational
 - In many settings, subjects would profit from increasing their depth of reasoning.
 - Use of the extensive form increases the willingness of subjects to believe in the rationality of their opponents.
- p-Beauty Contest data suggests subjects exhibit a great deal of heterogeneity in strategic thinking
 - Modal steps of iterated dominance is 2 or 3